

EDUGENIE

Google Gemini Powered Learning Assistant

6. Project Testing

Testing Objective

- Verify that each EduGenie feature accepts valid input, reaches the correct backend endpoint, and returns an understandable result.
- Check that the frontend and backend communicate correctly during local execution.

Functional Test Cases

- Q&A; Test – enter an educational question and verify that an answer is returned.
- Explanation Test – enter a topic and verify that a simplified explanation is produced.
- Quiz Test – provide educational text and verify generation of three questions with four options.
- Summary Test – provide a long paragraph and verify that a concise summary is returned.
- Learning Path Test – provide a topic and verify that a structured beginner-to-advanced recommendation is produced.

API Testing

- Verify the /qa endpoint.
- Verify the /explain endpoint.
- Verify the /quiz endpoint.
- Verify the /summarize endpoint.
- Verify the /learn/recommendations endpoint.

Integration Testing

- Start the application with Uvicorn.
- Open the local application at <http://127.0.0.1:8000>.
- Submit requests from the frontend and verify that responses appear in the result area.
- Check error handling when AI responses are invalid or incomplete.

Testing Outcome

- The project documentation identifies functional testing activities covering questions, explanations, quizzes, summaries, and personalized learning plans.

- Continuous testing and prompt refinement are identified as important for accuracy and usability.

Source basis: EduGenie project documentation provided by the user.